MIL-N-60189(MU)
AMENDMENT 4
12 July 1968
SUPERSEDING
AMENDMENT 3
9 June 1967

MILITARY SPECIFICATION

NETWORK D CIRCUIT BOARD: 8623286

This amendment forms a part of Military Specification MIL-N-60189(MU), dated 23 February 1965.

Paragraph 3.4.4.1: Delete and substitute:

"3.4.4.1 The dynamic requirements of 3.4.1.1 shall apply except that the simulated load of Figure 3 shall be used."

Paragraph 3.4.4.3(b): Delete "6.2 plus or minus 0.31 volts." and substitute "6.0 plus or minus 0.5 volts."

Following paragraph 4.1.1, add paragraph 4.1.2, 4.1.2.1 and 4.1.2.2.

"4.1.2 First Article (Initial Production) Approval.- The requirement for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of three (3) units manufactured in the same manner using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4.1.2.1 Government Testing.- When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to insure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for materials.

4.1.2.2 Contractor Testing.- When the contractor is responsible for conducting first article approval tests, the sample shall be inspected and a record of this inspection, including certificates of conformance for materials shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection."

Paragraph 4.2.3: Add the following sentence: "No lots shall be accepted until control samples have been satisfactorily tested."

* Paragraph 4.3.2 Delete first sentence and substitute:

"Units shall be selected at random by a government representative as control samples from each three (3) months production or portion thereof as follows:

FSC 1220

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3 units per first 100 produced
5 units per 200 "
7 units per 300 "
9 units per 400 "
10 units per 500 "
12 units per 700 "
15 units per 1000 "
and 2 units for each 100 items produced in excess of 1000."

Delete paragraph 4.5.1.3.2 and substitute the following:

"4.5.1.3.2 Operating.- This test shall be conducted in accordance with the humidity test defined in Procedure III of MIL-F-13926, except that the humidity shall be maintained at 92 to 97 percent for the 24-hour period, and the test shall cease at the end of that period (Method 106 of MIL-STD-202 shall be excluded). Throughout the test period, the board shall be capable of performing the tests of 4.5.3 shall be performed a minimum of once each 6 hours of the 24-hour period. The final test shall be conducted during the final portion of the 24-hour period. Between tests, the board shall be electrically de-energized.

Delete paragraph 4.5.1.4 and substitute the following:

"4.5.1.4 Salt Spray.- This test shall be conducted in accordance with the salt spray test defined in Method 101, Condition B of MIL-STD-202 using a 5 percent salt solution. At the end of the exposure period, salt deposits shall be removed from the board by a gentle wash, not exceeding 30 minutes, in running water not warmer than 100°F, using a soft-hair brush or plastic-bristle brush. The board shall then be air dried in a controlled environment at 72°F ± 5°F and a relative humidity of 30 to 50 percent for a period of 24 hours. The board shall be inspected visually for any physical damage, then tested in accordance with 4.5.2 for compliance with 3.3.6.

Paragraph 4.5.2.2.3(c): Delete "-6,2± 0.3 volts." and substitute "minus 6.0 plus or minus 0.5 volts."

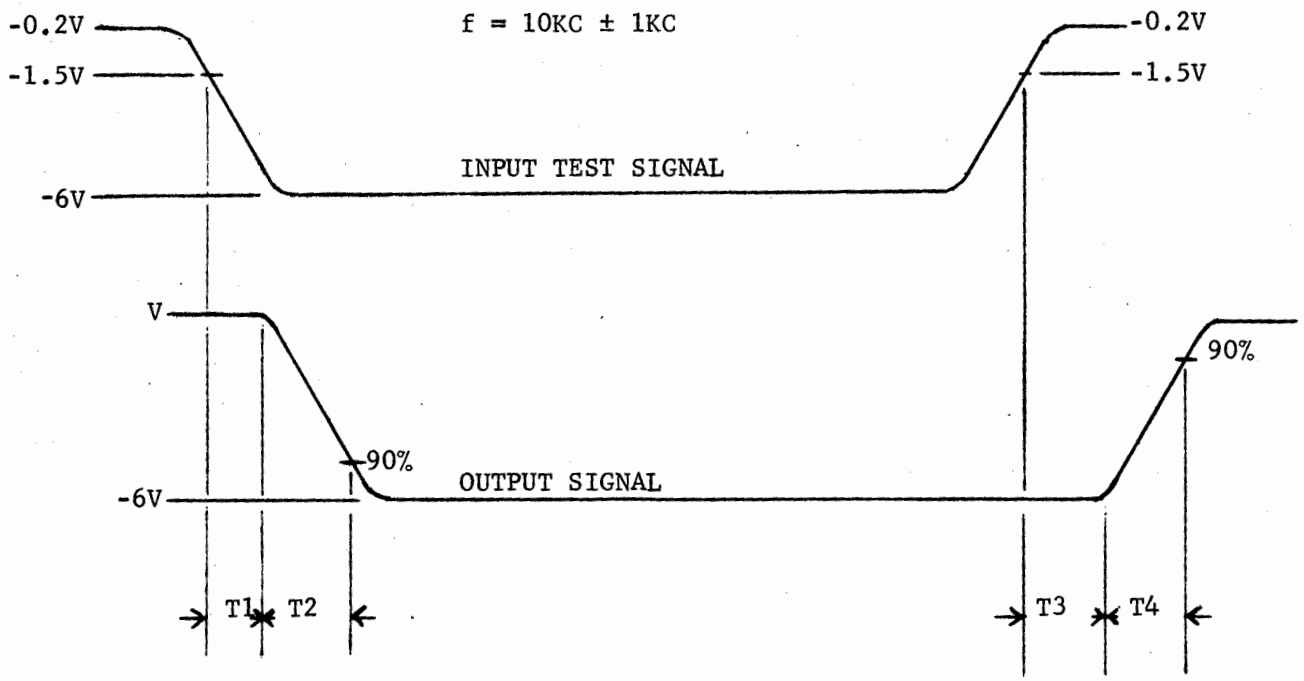
Paragraph 4.5.2.3.1(a): Delete "Table VI" and substitute "Table IX".

Paragraph 4.5.2.6(a): Delete "27000 ohms ± 1 percent." and substitute "27000 ohms ± 5 percent."

Paragraph 4.5.3.5(a): Delete "-6 ±0.5" and substitute "-6 ± 0.42".

Figure 2: Delete and substitute new Figure 2 shown of Page 3 of this amendment.

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Circuit	Output Volts (v)				
DUR. DLR.	+3 ± 0.12	T1 + T2 < 400 Nano. Sec.		T3 + T4	400 Nano. Sec.
REGULAR	-0.15 ± 0.15	T1 + T2 < 400 Nano. Sec.		T3 + T4	400 Nano. Sec.
		T1	T2	T3	T4
CLOCK	+6 ± 1.0	<3μ sec.	<1μsec.	<3μ sec.	< 500 nano sec.
CLOCK INHIBIT	+3 ± 0.12	<5μ sec.	<1.5μsec.	< 9μsec.	< 1.5μsec.

FIGURE 2

Delay time comparison of input and output waveforms. The delay time (T_1 and T_2) from the time the square wave applied to the input gate reaches minus 1.5 volts (going towards minus 6 volts) until the output reaches 90% of its total excursion (going towards minus 6 volts) and delay time (T_3 and T_4) measured from the time the square wave applied to the input gate reaches minus 1.5 volts (going towards zero volts) until the output reaches 90% of its total excursion (going towards the false state plus 3 volts) shall be as indicated in the table. The delay time (T_1) is measured from the time the square wave is applied to the input gate reaches minus 1.5V (going toward -6V) until output just starts descending and the delay time T_3 is measured from the time the square wave applied reaches -1.5V (going toward zero volts) until the output just starts rising. For extreme temperatures add 5% to the limits of $T_1 + T_2$ and $T_3 + T_4$.

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